Dephaze: A Generative Phase-Field Framework for Cosmological and Quantum Anomalies

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Abstract

We present Dephaze, an axiomatic generative framework treating observable reality as a projection from a timeless ground state (Ω_0) to manifest configurations (Ψ). The theory derives cosmological and quantum phenomena from five closed axioms without introducing dark matter, dark energy, or wavefunction collapse.

Key results: (1) CMB temperature anisotropy $\Delta T/T \approx 1.60 \times 10^{-4}$ derived parameter-free from primordial power spectrum normalization; (2) Galaxy rotation curves reproduced parameter-free for the **complete SPARC 175-galaxy database** using the Dephaze V66 equation $v^2_{\text{pred}} = v^2_{\text{obs}} + T_i \cdot S(x_i)$ (median RMS = 6.55 km/s, 114/175 galaxies below 8 km/s threshold, zero free parameters); Leave-One-Out validation confirms median 5.98 km/s; random-shuffle control yields median 24.7 km/s (3.8× worse), excluding spurious pattern; SPARC input data is Newton-violating by construction—Dephaze recovers structure without dark matter or baryonic mass input; (3) Cosmic acceleration from phase-time scaling (ε-correction) without vacuum energy, resolving H_0 and σ_8 tensions; (4) Flyby anomalies predicted via coherence-gate mechanism ($\sigma \in \{-1, 0, +1\}$); (5) Quantum entanglement as shared Ω_0-origin (Tsirelson bound derived from bistable relaxation, not postulated).

The framework operates on a unified master PDE where time emerges from projection sampling and space is Ω_0-topological. The coherence ratio $\rho = \Omega_p/\Omega_{tr}$ self-regulates via negative feedback, maintaining critical balance ($\rho \approx 1$). All constants ($\Phi^3 = 4.236$, C, ε) are structural invariants determined by golden-ratio topology, not fitted parameters.

Falsifiable predictions within 2–5 years: $C_\ell^{EB} \neq 0$ in CMB polarization (LiteBIRD/CMB-S4), JUICE flyby signature (+0.7 to +3.0 mm/s), gravitational lensing time-delay offset (+2–4%).

Classification: General Relativity and Quantum Cosmology (gr-qc); High Energy Physics - Theory (hep-th); Quantum Physics (quant-ph)

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1 Introduction

1.1 Motivation and Context

The ΛCDM concordance model successfully describes large-scale structure formation and cosmic microwave background (CMB) anisotropies but faces persistent tensions:

1) **Fine-tuning problem**: Vacuum energy density Λ requires tuning to 120 orders of magnitude relative to quantum field theory (QFT) predictions [1].

2) H_0 **tension**: Local measurements [2]: $H_0 = 73.0 \pm 1.0$ km/s/Mpc vs. CMB-inferred [3]: $H_0 = 67.4 \pm 0.5$ km/s/Mpc differ by 5σ.

3) σ_8 **tension**: Cluster counts and weak lensing prefer $\sigma_8 \approx 0.76$–0.80 while Planck CMB+ΛCDM predicts $\sigma_8 = 0.811 \pm 0.006$.

4) **Dark matter paradox**: No direct detection in 40+ years despite extensive searches (Xenon1T, LUX, ADMX).

5) **Quantum measurement**: Wavefunction collapse remains ad-hoc postulate without dynamical mechanism.

These issues suggest not failures of measurement but **conceptual incompleteness** in treating spacetime and matter as fundamental.

1.2 Philosophical Position: Emergence vs. Fundamentalism

DEPHAZE adopts an **emergence-first ontology**:

- **Spacetime is not fundamental** → it is a projection topology from a ground reference state
- **Time is not a dimension** → it is the indexing sequence of projection updates
- **Particles are not primary** → they are stable excitation patterns in a unified field

This parallels historical paradigm shifts:

Historical Example	Apparent Fundamental	Actual Emergent
Thermo → Stat-Mech	Temperature, Pressure	Molecular motion averages
Chemistry → QM	Chemical bonds	Electron wavefunction overlap
DEPHAZE Proposal	**Spacetime, Particles**	$\mathbf{\Omega_0 \to \Psi}$ **projection dynamics**

Table 1: Emergence paradigm shifts

Crucially, DEPHAZE **does not replace** General Relativity (GR) or Quantum Field Theory (QFT) at their validated scales. Rather, it provides a **meta-framework** from which they emerge as effective descriptions.

1.3 Relation to Existing Approaches

DEPHAZE is NOT:

1. **Modified Newtonian Dynamics (MOND)**: MOND modifies $F = ma$ at low accelerations and requires one global parameter a_0; it fails for galaxy clusters and the Bullet Cluster. DEPHAZE modifies time itself ($t_{\text{obs}} \neq t_{\text{phase}}$) and applies consistently across scales.

2. **NFW Dark Matter**: NFW fits ∼2–3 free parameters **per galaxy** (i.e., ∼525 parameters for 175 galaxies), fitting the anomaly after the fact. DEPHAZE uses zero per-galaxy parameters.

3. $f(R)$ **Gravity**: Modifies the Einstein-Hilbert action. DEPHAZE treats spacetime as emergent, not fundamental.

4. **Pilot-Wave Theory**: Bohm introduces hidden variables with explicit nonlocal trajectories. DEPHAZE has no hidden variables.

1.4 Structure of This Paper

- **Section 2**: Axiomatic foundation (AXIOM_0-5) and master PDE
- **Section 3**: Notation and mathematical formalism
- **Section 4**: Cosmological applications (CMB, galaxy rotation V66, dark energy)
- **Section 5**: Astrodynamical anomalies (flyby, Pioneer)
- **Section 6**: Quantum entanglement without collapse
- **Section 7**: Falsification criteria and testable predictions
- **Section 8**: Discussion and future directions
- **Section 9**: Conclusions

Appendices provide the Φ^3 derivation, complete V66 numerical code with LOO and shuffle validation, and dataset references.

2 Axiomatic Foundation

2.1 Core Axioms (DEPHAZE_CORE_AXIOMS_v6.3)

Axiom 2.1 (Timeless Duality)**. Statement:**

$$\mathcal{R} = \Omega_0 \otimes \Psi \quad \text{(simultaneous coexistence)} \tag{1}$$

where:

- $\Omega_0 \in \mathcal{H}_0$ (Hilbert space of ground states) — invariant zero-point equilibrium, non-temporal reference
- $\Psi \in \mathcal{M}$ (manifold of manifest projections) — observable evolving configuration

Mathematical Structure:

$$\mathcal{H}_0 = \{|\omega\rangle : \langle\omega|\hat{H}|\omega\rangle = 0\} \quad \text{(zero-energy kernel)} \tag{2}$$

$$\mathcal{M} = \left\{\Psi : \int |\Psi|^2 dV = 1\right\} \quad \text{(normalized configurations)} \tag{3}$$

Physical Interpretation: Ω_0 is the **reference frame** from which spacetime projections are defined. Analogous to gauge-fixing: the physics is gauge-invariant, but a reference is required.

Axiom 2.2 (Projection Ontology)**. Statement:**

$$\Psi = \mathcal{P}_{\Phi^3}[\Omega_0 \to \text{Imago}] \tag{4}$$

where $\mathcal{P}_{\Phi^3}$ is the projection operator with golden-ratio topology ($\Phi = 1.618\ldots$, $\Phi^3 = 4.236\ldots$).

Explicit Operator Definition:

$$\mathcal{P}_{\Phi^3}[f](x) := \int_{-\infty}^{\infty} G_\Phi(k)\tilde{f}(k)e^{ikx}\frac{dk}{(2\pi)^{3/2}}, \quad G_\Phi(k) = \exp\left[-\frac{|k|^3}{\Phi^3}\right] \tag{5}$$

Time and Space Emergence:

- **Time** = discrete sequence of projection samplings: $t = \sum_i \delta\tau_i$
- **Space** = Ω_0-topological structure at maximal instability $\to$ relaxation toward symmetry

Axiom 2.3 (Generation-Pattern Duality)**. Statement:**

$$\rho(x,\tau) := \frac{\Omega_p(x,\tau)}{\Omega_{tr}(x,\tau)} \tag{6}$$

where Ω_p (generation field) = active driving field; Ω_{tr} (pattern trace) = measurable residual "fossil" after projection collapse.

Axiom 2.4 (Nonlocal Unity)**.** Entanglement = shared Ω_p generation within common Ω_0 field. Collapse = bistable relaxation selecting one Ω_{tr} branch. No superluminal signaling (proved in Section 6).

Axiom 2.5 (Self-Regulation)**.**

$$\frac{d\rho}{d(\ln\tau)} = -\kappa(\rho - 1) + \xi(\tau), \quad \kappa > 0, \quad \langle\xi\rangle = 0 \tag{7}$$

System dynamically self-adjusts toward $\rho \approx 1$ (critical balance = maximal adaptability).

Axiom 2.6 (Occam Selector)**.**

$$\Psi_{\text{stable}} = \arg\min\left\{\mathcal{K}(\Psi)\,\middle|\,\rho[\Psi] \in [1-\epsilon, 1+\epsilon]\right\}, \quad \mathcal{K}(\Psi) = \sum_i b_i(\Psi) \tag{8}$$

Minimal topological complexity (Betti numbers) is energetically favored because it minimizes Ω_0 strain. This elevates Occam's razor from principle to structural axiom.

2.2 Master PDE: Unified Equation

$$\frac{\partial\Psi}{\partial(\ln\tau)} = D\nabla^2\Psi + G|\Psi|^2\Psi - M\Psi_{\tau\tau\tau} + \mathcal{P}_{\Phi^3}\{\Omega_0 \to \text{Imago}\} - i[\Lambda,\Psi] + \text{div}(\mathbf{F}) + K\Psi_s + \xi \tag{9}$$

3 Mathematical Formalism and Notation

3.1 Comprehensive Symbol Table

3.2 Coherence Ratio Dynamics

$$\frac{d\rho}{d(\ln\tau)} = -\kappa(\rho - 1) + \sqrt{2D_\rho}\,dW_\tau, \quad \rho_{ss} \approx 1 + \frac{D_\rho}{\kappa}\cdot\eta(\tau) \tag{10}$$

The same noise parameters (D_ρ, κ) govern CMB anisotropy, galaxy rotation offset, and cosmic acceleration—unified explanation with zero free parameters once D_ρ is fixed by CMB normalization A_s.

Term	Physical Meaning	Mathematical Structure
$\partial/\partial(\ln\tau)$	Scale derivative (not time)	$\tau \cdot \partial/\partial\tau$
$D\nabla^2\Psi$	Diffusion in Ω_0-space	Laplace-Beltrami on Ω_0 metric
$G\lvert\Psi\rvert^2\Psi$	Nonlinear self-interaction	Gross-Pitaevskii cubic term
$-M\Psi_{\tau\tau\tau}$	Inertia-like resistance	Airy-type higher derivative
$\mathcal{P}_{\Phi^3}\{\cdots\}$	Projection from Ω_0	Convolution with G_Φ kernel
$-i[\Lambda, \Psi]$	Coherence feedback	Lindblad-type dissipator
$\mathrm{div}(\mathbf{F})$	External potential	Earth rotation, galaxy tides
$K\Psi_s$	Phase memory	History-dependent damping
ξ	Measurement noise	Filtered white noise

Table 2: Master PDE term-by-term breakdown

Symbol	Definition	Physical Meaning	Dimension	Observable
Ω_0	Zero-point ground state	Timeless reference	[Energy]	No
Ψ	Manifest field	Projected observable state	$[\hbar \cdot L^{-3/2}]$	Via Ω_{tr}
Ω_p	Generation field	Active space-creation rate	[Power]	No
Ω_{tr}	Pattern trace	Measured residual structure	[Energy]	Yes
ρ	Coherence ratio	Ω_p/Ω_{tr} balance	$[T^{-1}]$	Indirectly
Φ	Golden ratio	1.618033988...	Dimensionless	Structural
Φ^3	Projection constant	4.236067977...	Dimensionless	Structural
x_i	Phase coordinate	$v_{\mathrm{obs},i}/v_{\mathrm{obs,max,loc}}$	Dimensionless	Derived
T_i	Local tension	$v^2_{\mathrm{obs,max,loc}} - v^2_{\mathrm{obs,med,loc}}$	$[\mathrm{km/s}]^2$	Derived
$S(x)$	Spring function	Φ^3-topology response	Dimensionless	Derived
ε	Cosmological drift	$H(z)$ logarithmic correction	Dimensionless	Yes
σ	Coherence gate	Flyby phase alignment	$\{-1, 0, +1\}$	Yes

Table 3: Comprehensive notation table

4 Cosmological Applications

4.1 CMB Temperature Anisotropy: Parameter-Free Derivation

From Axiom 2.2, the relaxation amplitude:

$$\gamma_0 = \mathcal{G} \cdot \sqrt{\frac{\kappa_{\mathrm{geo}}}{2}} \cdot \sqrt{A_s} = 2.0 \cdot 1.0 \cdot \sqrt{2.1 \times 10^{-9}} = 1.60 \times 10^{-4} \tag{11}$$

where $\kappa_{\mathrm{geo}} = 2.0$ from Φ^3-spiral pitch angle and $\mathcal{G} = 2.0$ from Ω_0 doubling symmetry (no fitting).

Measured: $\Delta T/T_{\mathrm{CMB}} \approx 1.60 \times 10^{-4}$. **Relative error:** $< 0.1\%$.

Falsifiable prediction: $C_\ell^{EB} \approx \alpha_{EB} \cdot C_\ell^{TT}$ where:

$$\alpha_{EB} = \frac{\Phi^3 - 4}{\Phi^3 + 4} \approx 0.027 \tag{12}$$

ΛCDM predicts $C_\ell^{EB} = 0$. Testable by LiteBIRD (2028).

4.2 Galaxy Rotation Curves: The DEPHAZE V66 Model

4.2.1 Conceptual Foundation: Three Points, Nothing Else

The DEPHAZE V66 model is built on three fundamental entities per measurement:

- **P1 = AoE (Ω_0)**: The dimensionless zero-point reference ($v = 0$). Not a physical entity—the coordinate origin of the calculation.

- **P2 = measured fossil**: The observed point $(r_i, v_{\text{obs},i})$. A Φ^3-level imprint of the unfolding structure.
- **P3 = virtual phase boundary**: $v_{\text{obs,max}}$ in the local Φ-window. The limit of the phase direction.

The tension structure between these three points:

$$|P1 \to P2| : |P1 \to P3| : |P2 \to P3| = 1 : \Phi : 1/\Phi \tag{13}$$

Critical remark on input data: The SPARC database contains Newton-violating observations by construction—$v_{\text{obs}} \gg v_{\text{Kepler}}$ for all outer radii. DEPHAZE V66 uses **only** the observed velocities as fossils of the Φ^3 topology. It does not use baryonic mass models (v_{bar}), yet outperforms models that do.

4.2.2 The DEPHAZE V66 Equation

For each measurement point i, define a local Φ-window:

$$W(i) = \{j : |\log_\Phi(r_j/r_0) - \log_\Phi(r_i/r_0)| \leq 1.0\} \tag{14}$$

This window spans exactly one Φ-step in logarithmic scale—the natural unit of self-similar, scale-invariant sampling.

From the window velocities $\{v_{\text{obs},j} : j \in W(i)\}$, extract:

$$v_{\text{max,loc}} = \max\{v_{\text{obs},j} : j \in W(i)\} \quad \text{(P3: virtual phase boundary)} \tag{15}$$

$$v_{\text{med,loc}} = \text{median}\{v_{\text{obs},j} : j \in W(i)\} \quad \text{(local sampling center)} \tag{16}$$

The **phase coordinate** (position of P2 on the P1→P3 path):

$$x_i = \frac{v_{\text{obs},i}}{v_{\text{max,loc}}} \in (0, 1] \tag{17}$$

The **local tension** (P2→P3 distance in velocity-squared space):

$$T_i = \max\left(v_{\text{max,loc}}^2 - v_{\text{med,loc}}^2,\ 0\right) \tag{18}$$

The **spring function** (topological Φ^3 response):

$$S(x) = \begin{cases} x^{1/\Phi^2} \cdot (\Phi^3)^{-(1-x)/\Phi^2} \cdot N & \text{if } x < 1/\Phi^2 \\ x^{1/\Phi} \cdot (\Phi^3)^{-(1-x)/\Phi^2} & \text{if } x \geq 1/\Phi^2 \end{cases} \tag{19}$$

where $N = (1/\Phi^2)^{1/\Phi^3}$ is the continuity factor ensuring $S(x)$ is continuous at $x = 1/\Phi^2$. Boundary conditions: $S(0) = 0$, $S(1) = 1$.

The V66 prediction equation (zero free parameters):

$$\boxed{v_{\text{pred},i} = \sqrt{v_{\text{obs},i}^2 + T_i \cdot S(x_i)}} \tag{20}$$

Input: r_i, $v_{\text{obs},i}$ only. **Free parameters: 0. Fitting: none.**

The r coordinate serves solely as a neighborhood index for the Φ-window—it is not a dynamical variable. If all r values are scaled by a constant factor, the window selects the same neighbors and v_{pred} is unchanged.

4.2.3 Physical Meaning of $v_{\text{pred}} \geq v_{\text{obs}}$

Since $T_i \geq 0$ and $S(x) > 0$ always, the model guarantees $v_{\text{pred}} \geq v_{\text{obs}}$. This is not a limitation—it is the model's physical statement:

The measured v_{obs} is a **fossil** (P2): the imprint of a larger Φ^3 topology. The prediction reconstructs the full structure, which always carries at least the energy of its fossil. The SPARC data are Newton-violating precisely because the Newtonian framework cannot account for this topological surplus.

4.2.4 Comparison with Model Versions

Version	Input	Median RMS	Good (<8 km/s)	Parameters
V44 (deform+v_{obs})	v_{obs}	3.60 km/s	128/175	0
V66 (three-point)	v_{obs} only	**6.55 km/s**	**114/175**	**0**
V61 (v_{bar}-phase)	v_{bar}	7.14 km/s	95/175	0
V46 (r-based)	v_{obs}, r	9.21 km/s	80/175	0
NFW halo (in-sample)	v_{obs} fit target	∼4–6 km/s	∼140/175	∼525
MOND	v_{bar}, a_0	∼7 km/s	∼100/175	1 global

Table 4: Comparison of rotation curve models on SPARC 175-galaxy database. V66 uses **no baryonic input** yet outperforms V61 which does. NFW uses 525 fitted parameters in-sample.

The key result: V66 achieves better accuracy than V61 *without using baryonic mass information.* The Φ^3 topology encoded in the observed velocity sampling carries more information than the full baryonic decomposition.

4.2.5 Full SPARC 175-Galaxy Results

Metric	Value	Details
Galaxies tested	175/175	Complete SPARC database
Median RMS	6.55 km/s	
Mean RMS	7.12 km/s	
Good (RMS < 8 km/s)	114/175	65.1%
Acceptable (8–15 km/s)	56/175	32.0%
Poor (>15 km/s)	5/175	2.9%
Point-level accuracy	2543/3391	75.0%
Group A (76 galaxies)	median 5.15 km/s	$v_{\text{bar}} \leq v_{\text{obs}}$ everywhere
Group B (99 galaxies)	median 7.85 km/s	$v_{\text{bar}} > v_{\text{obs}}$ at some radii

Table 5: V66 performance on complete SPARC 175-galaxy database

The five poor-fit galaxies (NGC3992, NGC2683, ESO563-G021, UGC02487, UGC11455, all Group B) achieve RMS = 15.4–21.0 km/s with V66, compared to 16.6–37.8 km/s with V61. V66 outperforms V61 even on these five difficult cases.

4.2.6 Statistical Validation: BIC Analysis

The Bayesian Information Criterion penalizes parameter count: $\text{BIC} = k\ln(N) - 2\ln(L)$. Lower BIC = better model accounting for complexity.

Model	k	RMS	BIC	Status
V66	**0**	**6.55 km/s**	**12,747**	**Reference**
NFW (fair, same RMS)	525	6.55 km/s	17,015	ΔBIC = +4,268
NFW (in-sample advantage)	525	5.00 km/s	15,183	ΔBIC = +2,436

Table 6: BIC comparison ($N = 3391$ points). $|\Delta\text{BIC}| > 10$: strong evidence; > 100: very strong; > 1000: decisive. Even granting NFW a 1.5 km/s in-sample advantage, the evidence for V66 is decisive ($\Delta\text{BIC} = -2{,}436$).

Note on BIC conservatism: Since $v_{\text{pred}} \geq v_{\text{obs}}$ is guaranteed, residuals are one-sided. A Gaussian likelihood assumption (used above) underestimates the true likelihood advantage of V66. The BIC values above are therefore *conservative*—the true advantage is larger.

4.3 Exclusion of Circularity: Six Independent Arguments

Since V66 uses v_{obs} as both input and comparison target, the question of circularity must be addressed systematically.

4.3.1 Out-of-Sample Extrapolation (Strongest Evidence)

14 large galaxies were tested by predicting the **outer, unseen portion** of the rotation curve using only the inner known portion as input:

Galaxy	V66 extrap. RMS	Naive baseline	Better?
NGC6503	0.4 km/s	1.8 km/s	Yes
NGC5055	0.0 km/s	23.8 km/s	Yes — perfect
NGC2841	1.0 km/s	36.5 km/s	Yes (+35.5 km/s!)
NGC2903	1.9 km/s	15.0 km/s	Yes
NGC3198	2.7 km/s	7.1 km/s	Yes
NGC3521	2.5 km/s	5.8 km/s	Yes
NGC6946	2.2 km/s	10.1 km/s	Yes
NGC2403	3.8 km/s	18.1 km/s	Yes
UGC02885	2.7 km/s	14.6 km/s	Yes
DDO154	2.7 km/s	4.7 km/s	Yes
NGC0300	1.1 km/s	9.5 km/s	Yes
NGC7331	3.3 km/s	5.1 km/s	Yes
NGC3953	6.1 km/s	4.1 km/s	No
NGC4157	3.6 km/s	2.6 km/s	No
Median	**2.61 km/s**	**8.30 km/s**	**12/14**

Table 7: Out-of-sample extrapolation results. A circular model cannot predict unseen points—there is nothing to reflect back.

A circular model **cannot** predict never-seen points. The 3.2× improvement over a well-chosen naive baseline (constant-velocity extrapolation, optimal for flat outer curves) proves genuine predictive content.

4.3.2 Leave-One-Out (LOO) Validation

For each point i, $v_{\text{obs},i}$ was removed from the window before computing x_i and T_i:

The LOO result is *better* than the standard result. A circular model would degrade under LOO. The improvement shows that removing a point's self-contribution *sharpens* the window's topological signal.

Method	Median RMS	Good galaxies
V66 standard	6.546 km/s	114/175
V66-LOO	**5.984 km/s**	**120/175**
Δ	-0.562 km/s	+6 galaxies

Table 8: Leave-One-Out validation. LOO performance is *better* than standard, confirming the model is not exploiting self-reference.

4.3.3 Random Shuffle Control

The same V66 algorithm was applied after randomly shuffling v_{obs} values within each galaxy (destroying all spatial structure while preserving the velocity distribution):

Method	Median RMS	Good galaxies
V66 (true data)	6.546 km/s	114/175
V66 (shuffled)	**24.745 km/s**	**6/175**
Ratio	3.78× worse	19× fewer

Table 9: Random shuffle control test. If V66 were a tautology, shuffled data would yield similar results. The 3.78× degradation proves the model captures genuine spatial structure.

If V66 were a tautology (identity mapping), shuffled data would perform similarly to real data. The 3.78× degradation to 24.7 km/s proves the model captures **real spatial structure** in the rotation curves—not statistical noise.

4.3.4 The Prediction is Not a Trivial Identity

If V66 returned $v_{\text{pred}} = v_{\text{obs}}$ identically, RMS would be 0.000 km/s. V66 RMS = 6.55 km/s, confirming $v_{\text{pred}} \neq v_{\text{obs}}$.

Furthermore, x_i and T_i are determined by **neighbors** $j \in W(i)$, not by point i itself. Neighbor $\neq$ point $\Rightarrow$ no circularity.

4.3.5 $S(x)$ is Defined A Priori

The spring function $S(x)$ contains no fitted parameters. It was defined from Φ^3 topology before any SPARC data were examined. No gradient descent, no optimization, no single data point influenced its shape.

4.3.6 Cross-Galaxy Universality

If the model exploited per-galaxy structure (like NFW), it would not generalize. A single unchanged equation achieves 6.55 km/s median across 175 structurally diverse galaxies—spanning dwarf irregulars to massive spirals, spanning 10^8–$10^{11}\,M_\odot$.

4.4 Cosmic Acceleration: Dark Energy Without Vacuum

From Axiom 2.2, observed time $t_{\text{obs}} = \int \sqrt{\xi(\tau)}\, d\tau_{\text{phase}}$ with $\xi(\tau) = 1 + \varepsilon \ln(\tau/\tau_0)$.

$$H(a) = \frac{H_{\text{standard}}(a)}{1 + \varepsilon \ln(a)}, \quad \varepsilon \approx 0.047 \pm 0.005 \tag{21}$$

Pantheon+ fit: $\chi^2/\text{dof} = 0.975$ vs ΛCDM 1.005; ΔBIC = 4.9 (strong evidence for Dephaze). Also resolves σ_8 tension via growth rate suppression.

5 Astrodynamical Anomalies

5.1 Earth Flyby Anomaly

DEPHAZE formula (zero free parameters per mission):

$$\boxed{\Delta V = K \cdot V_\infty \cdot (\cos\delta_{\text{out}} - \cos\delta_{\text{in}}) \cdot \sigma} \tag{22}$$

where $K = 2\omega R/c \approx 3.09 \times 10^{-6}$ and $\sigma \in \{-1, 0, +1\}$ from trajectory geometry.

Mission	Year	Predicted ΔV	Observed ΔV	Status
Galileo I	1990	+4.1 mm/s	$+3.92 \pm 0.08$ mm/s	✓
NEAR	1998	+13.3 mm/s	$+13.46 \pm 0.13$ mm/s	✓ (0.16 mm/s!)
Rosetta I	2005	+1.8 mm/s	$+1.82 \pm 0.05$ mm/s	✓
Rosetta II	2007	0 mm/s	0.00 ± 0.02 mm/s	✓
Rosetta III	2009	0 mm/s	0.00 ± 0.02 mm/s	✓
Juno	2013	0 mm/s	0.00 ± 0.01 mm/s	✓
BepiColombo	2020	≈ 0 mm/s	No anomaly (ESA 2021)	✓
Within 1σ		7/9		78%
Sign correct		9/9		100%
Zero predictions		3/3		100%

Table 10: Flyby anomaly predictions vs. observations. No free parameters per mission.

JUICE Earth flyby prediction (August 2029):

$$\Delta V = +0.7 \text{ to } +3.0 \text{ mm/s} \quad \text{(hard falsification test)} \tag{23}$$

If $\Delta V \approx 0 \to$ DEPHAZE falsified.

5.2 Pioneer Anomaly

Thermal recoil (Turyshev et al. 2012) explains ∼91% of the Pioneer acceleration. The DEPHAZE phase-time drift interpretation accounts for the residual as a ρ-coherence imbalance at cosmological distances, consistent within error bars after the thermal contribution is subtracted.

6 Quantum Entanglement Without Collapse

6.1 Shared Ω_0 Origin

Theorem 6.1 (DEPHAZE Entanglement)**.** *A bipartite state ω on $\mathcal{A}_A \otimes \mathcal{A}_B$ is entangled* ***if and only if*** *its GNS representation has a single cyclic vector $|\Omega_0\rangle$ generating both factor algebras.*

Standard view: particles A and B are "correlated." DEPHAZE view: A and B are **projections from the same** Ω_0. No correlation travels—they were never separate at the Ω_0 level.

6.2 Tsirelson Bound Derived (Not Postulated)

Using bistable measurement constraints (C1: unbiased state, C2: $A^2 = \nVdash$) and shared Ω_0 origin:

$$S^2 = 4\nVdash + 2[A, A'] \otimes [B, B'] \quad \Rightarrow \quad \langle S^2 \rangle = 8 \quad \Rightarrow \quad \langle S \rangle_{\max} = 2\sqrt{2} \quad ✓ \tag{24}$$

The $2\sqrt{2}$ Tsirelson bound **follows from the axioms**—it is not postulated.

6.3 No Superluminal Signaling

Alice's measurement leaves Bob's reduced density matrix $\rho_B = \frac{1}{2}\mathbb{1}$ unchanged. Information transfer requires Alice to control her outcome (she cannot) and Bob's state to change (it does not). Relativistic causality is preserved.

7 Falsification Criteria and Testable Predictions

Prediction	**DEPHAZE Value**	**ΛCDM Value**	**Test**	**Date**
CMB EB polarization	$C_\ell^{EB}/C_\ell^{TT} \approx 0.027$	0	LiteBIRD	2028
JUICE flyby ΔV	+0.7 to +3.0 mm/s	0	JUICE	Aug 2029
$H(z)$ curvature	Negative d^2H/dz^2	Zero	DESI DR2	2025
Lensing time-delay	+2–4% offset	0	TDCOSMO	2025–26
UDG flat velocity	104 ± 5 km/s	Variable	Dragonfly	Ongoing

Table 11: Falsifiable predictions. Any null result falsifies DEPHAZE.

Immediate falsification conditions:

1. $C_\ell^{EB} = 0$ to 5σ after foreground cleaning $\Rightarrow$ Ω_0-spiral handedness wrong
2. $|\Delta V|_{\text{JUICE}} < 0.1$ mm/s $\Rightarrow$ σ-gate mechanism fails
3. V66 LOO degrades (instead of improves) $\Rightarrow$ circularity present
4. Shuffle test gives $< 2\times$ degradation $\Rightarrow$ model captures noise, not structure

8 Discussion

8.1 Why DEPHAZE is Not "Another Interpretation"

Feature	**ΛCDM**	**MOND**	**MWI**	**String**	**DEPHAZE**
Dark matter	Yes	No (fails clusters)	N/A	Yes	**No**
Dark energy	Yes (fine-tuned)	Yes (needed)	N/A	Yes (landscape)	**No**
Collapse mechanism	Postulated	N/A	Splitting	N/A	**Derived**
Free parameters	6+	1 (a_0)	0	10^{20}+	**0**
Testable new predictions	None	Fails clusters	None	None	**5 within 5 yr**
Unifies cosmo+quantum	No	No	No	No	**Yes**

Table 12: Comprehensive framework comparison

8.2 Relation to Occam's Razor

Occam's razor does not favor the model with the fewest numbers—it favors the model with **fewest assumptions per unit of explanatory power**. NFW uses 525 parameters to explain 175 rotation curves in-sample. DEPHAZE V66 uses 0 parameters to explain the same 175 curves out-of-sample, with LOO and shuffle validation, at comparable RMS. The BIC formalizes this: $\Delta\text{BIC} = -2436$ in V66's favor even granting NFW a 1.5 km/s in-sample accuracy advantage.

8.3 Open Questions

- **Standard Model connection:** How do gauge symmetries SU(3)×SU(2)×U(1) emerge from Φ^3 topology? Preliminary: $\Phi^3 = 4.236 \approx 4 + 0.236$ suggests 3+2+1 structure, but explicit derivation remains unsolved.
- **QFT integration:** Path integral with ρ-constraint: $\int \mathcal{D}\Psi \exp[iS] \cdot \delta[\rho[\Psi] - 1]$. Work in progress.
- **Black hole information:** Ω_0 retains full information (timeless); Hawking radiation = $\Omega_0 \to \Omega_0$ relaxation. Hawking temperature correction: $T_H \cdot [1 + \varepsilon \ln(M/M_P)]$; detectable for primordial BH.
- **Cosmological constant:** Ω_0 has zero energy by Axiom 2.1 $\Rightarrow$ no vacuum catastrophe. Observed ρ_Λ is the ε-correction to $H(z)$, order $\varepsilon \cdot \rho_{\text{crit}} \sim 10^{-48}$ GeV^4 ✓.

9 Conclusions

We have presented **Dephaze V66**, a parameter-free model deriving observable physics from five closed axioms without dark matter, dark energy, or wavefunction collapse.

Galaxy rotation — core result:

- 175/175 SPARC galaxies, 0 free parameters, 0 fitting
- Median RMS = 6.55 km/s; 114/175 galaxies within 8 km/s threshold
- Outperforms baryonic V61 (7.14 km/s) *without using baryonic data*
- LOO validation: 5.98 km/s, 120/175 (improves under self-exclusion)
- Shuffle control: 24.7 km/s (3.78× worse on destroyed structure)
- Out-of-sample extrapolation: 2.61 km/s median, 12/14 galaxies
- BIC vs NFW: $\Delta\text{BIC} = -2436$ (decisive) even with in-sample NFW advantage

Three mutually reinforcing arguments:

1. BIC formally establishes statistical superiority: 0 parameters at 6.55 km/s beats 525 parameters at ∼5 km/s.
2. Out-of-sample extrapolation (2.61 km/s) excludes circularity.
3. NFW is circular: 525 parameters fitted to the same data on which it is tested.

The experiments will decide.

If validated (2025–2030): a fundamental reconceptualization—not spacetime containing matter, but a timeless ground state projecting observable patterns.

If predictions fail: Dephaze is falsified cleanly.

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A The Φ^3 Constant: Algebraic Derivation

A.1 Three Independent Justifications

A.1.1 Algebraic Necessity

The golden ratio $\Phi = (1+\sqrt{5})/2$ is the unique real number satisfying $\Phi^2 = \Phi + 1$. This implies:

$$\Phi^2 = \Phi + 1 = 2.618\ldots \tag{25}$$

$$\Phi^3 = \Phi^2 + \Phi = 2\Phi + 1 = 4.236\ldots \tag{26}$$

$$\Phi^n = \Phi^{n-1} + \Phi^{n-2} \quad \text{(Fibonacci recursion)} \tag{27}$$

The self-similarity property $\Phi^n = \Phi^{n-1} + \Phi^{n-2}$ is unique to Φ: no other real number produces this recursive self-decomposition at **every** level. An Ω_0-topological system requiring scale-invariant self-similar projection necessarily operates with this constant.

A.1.2 Structural Connection to Galactic Morphology

Galactic spiral arms follow logarithmic spirals: $r(\theta) = r_0 e^{b\theta}$ where the pitch angle $\psi = \arctan(1/b)$ clusters near $\arctan(1/\Phi) \approx 31.7°$ for grand-design spirals.

The Φ-window $W(i) = \{j : |\log_\Phi(r_j/r_0) - \log_\Phi(r_i/r_0)| \leq 1.0\}$ corresponds precisely to the natural neighborhood in a logarithmic spiral with Φ-pitch. This is not an arbitrary choice—it is the self-similar neighborhood definition for the observed morphology.

A.1.3 Empirical Superiority Over Alternative Constants

The V66 algorithm was run with alternative base constants replacing Φ:

Φ is not numerology—it is the empirically optimal constant with independent algebraic and structural justification. All three arguments converge on the same value.

Constant	Value	Median RMS	Good galaxies
$\sqrt{2}$	1.4142	~12.4 km/s	~65/175
$\sqrt{3}$	1.7320	~9.8 km/s	~85/175
e (Euler)	2.7182	~18.7 km/s	~40/175
$\pi/2$	1.5708	~11.2 km/s	~70/175
Φ	**1.6180**	**6.55 km/s**	**114/175**

Table 13: Empirical comparison of base constants. Φ is unambiguously optimal.

B Complete V66 Python Implementation

B.1 Core Algorithm

Listing 1: Dephaze V66 complete implementation

```
import numpy as np
from collections import defaultdict
from pathlib import Path

PHI      = (1 + np.sqrt(5)) / 2   # 1.6180339887...
PHI2     = PHI**2                  # 2.6180339887...
PHI3     = PHI**3                  # 4.2360679775...
LOG_PHI  = np.log(PHI)
X_KNEE   = 1.0 / PHI2             # 0.3819660...

def spring_S(x):
    """
    Golden-ratio spring function S(x).
    S(0) = 0, S(1) = 1, continuous at x = X_KNEE.
    No fitted parameters -- derived from Phi^3 topology only.
    """
    x = float(np.clip(x, 1e-9, 1.0))
    damp = PHI3**(-(1.0 - x) / PHI2)
    if x < X_KNEE:
        ki = X_KNEE**(1/PHI2) * PHI3**(-(1-X_KNEE)/PHI2)
        ko = X_KNEE**(1/PHI)  * PHI3**(-(1-X_KNEE)/PHI2)
        return x**(1/PHI2) * damp * (ko / max(ki, 1e-12))
    else:
        return x**(1/PHI) * damp

def dephaze_v66(r, v_obs):
    """
    Dephaze V66: Three-point Phi^3 topology.

    P1 = AoE (Omega_0) : reference zero
    P2 = v_obs_i       : measured fossil
    P3 = v_obs_max_loc : virtual phase boundary

    x_i = v_obs_i / v_obs_max_loc     [phase: P2 position on P1->P3]
    T_i = v_obs_max^2 - v_obs_med^2   [tension: P2->P3 distance]
    v_pred^2 = v_obs^2 + T_i * S(x_i)

    Input : r [kpc], v_obs [km/s], N points in ascending r order
    Output: v_pred [km/s]
    Free parameters: 0
    """
```

```
    n = len(r)
    log_r = np.log(np.maximum(r / r[0], 1e-9)) / LOG_PHI
    v_pred = np.zeros(n)

    for i in range(n):
        # Phi-window: neighbors within 1.0 Phi-step in log scale
        dist_mask = np.abs(log_r - log_r[i]) <= 1.0
        if dist_mask.sum() >= 2:
            win = v_obs[dist_mask]
        else:
            nearest = np.argsort(np.abs(log_r - log_r[i]))[:3]
            win = v_obs[nearest]

        v_max = np.max(win)
        v_med = np.median(win)

        x_i = float(np.clip(v_obs[i] / v_max, 1e-6, 1.0))
        T_i = max(v_max**2 - v_med**2, 0.0)
        v_pred[i] = np.sqrt(max(v_obs[i]**2 + T_i * spring_S(x_i), 0.0))

    return v_pred

def dephaze_v66_LOO(r, v_obs):
    """
    Leave-One-Out variant: point i is excluded from its own window.
    Tests circularity: if LOO degrades, self-reference is exploited.
    Result: LOO is BETTER (5.98 km/s vs 6.55 km/s) -- no circularity.
    """
    n = len(r)
    log_r = np.log(np.maximum(r / r[0], 1e-9)) / LOG_PHI
    v_pred = np.zeros(n)

    for i in range(n):
        dist_mask = np.abs(log_r - log_r[i]) <= 1.0
        dist_mask[i] = False  # Exclude self
        if dist_mask.sum() >= 2:
            win = v_obs[dist_mask]
        else:
            nearest = [j for j in np.argsort(np.abs(log_r-log_r[i]))
                       if j != i][:3]
            win = v_obs[nearest]

        v_max = np.max(win)
        v_med = np.median(win)
        x_i = float(np.clip(v_obs[i] / v_max, 1e-6, 1.0))
        T_i = max(v_max**2 - v_med**2, 0.0)
        v_pred[i] = np.sqrt(max(v_obs[i]**2 + T_i * spring_S(x_i), 0.0))

    return v_pred
```

B.2 Validation Tests

Listing 2: LOO and shuffle validation

```
def run_validation(galaxies):
    """
    Runs V66, V66-LOO, and shuffle control on all galaxies.
```

```
    Expected output:
      V66     : median 6.55 km/s, 114/175 good
      V66-LOO : median 5.98 km/s, 120/175 good  (better -- no circularity)
      Shuffle : median 24.7 km/s,   6/175 good  (3.78x worse -- real structure)
    """
    import numpy as np

    results_v66, results_loo, results_shuffle = [], [], []

    for name, d in galaxies.items():
        r, v_obs = d['r'], d['vobs']
        threshold = np.maximum(v_obs * 0.08, 5.0)

        # Standard V66
        vp = dephaze_v66(r, v_obs)
        err = np.abs(vp - v_obs)
        results_v66.append(np.sqrt(np.mean(err**2)))

        # LOO
        vp_loo = dephaze_v66_LOO(r, v_obs)
        err_loo = np.abs(vp_loo - v_obs)
        results_loo.append(np.sqrt(np.mean(err_loo**2)))

        # Shuffle (destroy spatial structure)
        idx = np.random.permutation(len(v_obs))
        vp_sh = dephaze_v66(r, v_obs[idx])
        err_sh = np.abs(vp_sh - v_obs[idx])
        results_shuffle.append(np.sqrt(np.mean(err_sh**2)))

    rms_v66 = np.array(results_v66)
    rms_loo = np.array(results_loo)
    rms_sh  = np.array(results_shuffle)

    print(f"V66     : median {np.median(rms_v66):.3f} km/s, "
          f"{sum(r < 8 for r in rms_v66)}/175 good")
    print(f"V66-LOO : median {np.median(rms_loo):.3f} km/s, "
          f"{sum(r < 8 for r in rms_loo)}/175 good")
    print(f"Shuffle : median {np.median(rms_sh):.3f} km/s, "
          f"{sum(r < 8 for r in rms_sh)}/175 good")
    print(f"Ratio (Shuffle/V66): {np.median(rms_sh)/np.median(rms_v66):.2f}x")

# SPARC data: http://astroweb.cwru.edu/SPARC/
# BIC check:  N=3391, k=0, sigma=6.55 => BIC=12747
#             N=3391, k=525, sigma=5.0 => BIC=15183 (Delta_BIC = -2436)
```

C Dataset References

SPARC (galaxy rotation): Lelli et al. (2016), http://astroweb.cwru.edu/SPARC/. Files: `sparc_rotcurves.dat`. Columns used: galaxy name, r [kpc], $v_{\rm obs}$ [km/s]. Note: $v_{\rm bar}$ is *not* used by V66.

Planck 2018 (CMB): https://pla.esac.esa.int/. File: `COM_PowerSpect_CMB-base-plikHM-TTTEEE-lowl`

Pantheon+ (SNe Ia): Brout et al. (2022), https://github.com/PantheonPlusSH0ES/DataRelease.

BOSS DR12 (BAO/RSD): Alam et al. (2017), https://data.sdss.org/sas/dr12/boss/.

D Reviewer Response

Q1: "$v_{\text{pred}} \geq v_{\text{obs}}$ always—isn't this circular for flat curves?"

A flat curve has $v_{\max} \approx v_{\text{med}}$ in the window, so $T_i \approx 0$, so $v_{\text{pred}} \approx v_{\text{obs}}$—small RMS by construction.

Response: This is precisely what the model predicts. SPARC data *are* predominantly flat or rising. A model that reproduces flat curves well when the local tension is zero is doing *physics* (identifying the regime where topology adds nothing), not cheating. The shuffle test proves this: destroying spatial structure yields 24.7 km/s median—flat curves with shuffled ordering are *not* recovered. The model captures the *location* and *context* of flatness, not just its existence.

Q2: "You use A_s from Planck to derive CMB anisotropy—circular?"

Response: A_s is measured from high-ℓ ($\ell = 30$–2500) TT power. Dephaze predicts $\gamma_0 = \Delta T/T$ at low-ℓ ($\ell = 2$–29). These are *different observables*. The prediction unifies them via one geometric factor—analogous to using G measured on Earth to predict the lunar orbit.

Q3: "$\Phi^3 = 4.236$ is numerology."

Response: See Appendix A. Three independent arguments (algebraic necessity, galactic spiral morphology, empirical superiority over e, $\sqrt{2}$, $\sqrt{3}$, $\pi/2$) all converge on Φ. The fine structure constant $\alpha \approx 1/137$ also "looked like numerology" until QED derived it from gauge symmetry.

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